TEXORA · LECTURE SHEET

Infrared (IR) and Raman Spectroscopy

Reading molecules by their vibrations: the theory of infrared absorption, stretching and bending modes, what shifts a group frequency, how samples are handled and spectra recorded, the correlation charts that turn bands into structures, and Raman scattering — the complementary technique — with the applications of both in textile analysis.

TE-2205 · TEXTILE CHEMISTRY & COLORATION-I · TOPIC 7 · LEVEL 2 TERM 2

1. Theory: molecules as vibrating springs

Chemical bonds are not rigid rods: atoms vibrate about their equilibrium positions — stretching along the bond and bending away from it. These vibrations have natural frequencies in the **infrared region** of the electromagnetic spectrum. When IR radiation of exactly a vibration's frequency falls on the molecule, the energy is absorbed and the vibration's amplitude increases — provided the vibration changes the molecule's **dipole moment** (the selection rule for IR activity). An IR spectrum plots absorption against **wavenumber** $\tilde{\nu}$(cm^{-1}):

$$\tilde{\nu} = 1 \div \lambda \text{ (in cm)} = \nu \div c$$

WORKED EXAMPLE

Radiation of wavelength 2.5 μm: $\lambda = 2.5 \times 10^{-4}$ cm, so $\tilde{\nu} = 1 \div (2.5 \times 10^{-4}) =$ **4000 cm^{-1}** — the top of the routine mid-IR range (4000–400 cm^{-1}, i.e. 2.5–25 μm). Below ≈ 1500 cm^{-1} lies the **fingerprint region**, where the pattern is unique to each compound; above it, the **functional-group region** identifies particular bonds.

Treating a bond as a spring gives **Hooke's law** for the vibration frequency:

$$\tilde{\nu} = (1 \div 2\pi c) \times \sqrt{(k \div \mu)}, \quad \text{where } \mu = m_1 m_2 \div (m_1 + m_2)$$

k is the bond force constant (stiffness: single ≈ 500, double ≈ 1000, triple ≈ 1500 N/m in round numbers) and μ the reduced mass of the two atoms.

WORKED EXAMPLE — THE C–H STRETCH

For C–H: $\mu = (12 \times 1.008) \div 13.008 \approx 0.93$ amu $= 1.544 \times 10^{-27}$ kg; $k \approx 500$ N/m.
$\sqrt{(k/\mu)} = \sqrt{(500 \div 1.544 \times 10^{-27})} \approx 5.69 \times 10^{14}$ s^{-1}; $\nu =$ that $\div 2\pi \approx 9.06 \times 10^{13}$ Hz; $\tilde{\nu} = \nu/c \approx 9.06 \times 10^{13} \div 2.998 \times 10^{10} \approx$ **3020 cm^{-1}** — exactly where C–H stretching bands are observed (2850–3100

cm⁻¹). Two lessons follow: stronger bonds (larger k) vibrate higher, and heavier atoms (larger μ) vibrate lower.

2. Modes of vibration: stretching and bending

A molecule of N atoms has 3N − 6 fundamental vibrations (3N − 5 if linear). They divide into:

- **Stretching** — bond lengths change, angles do not: **symmetric** (both bonds of a group lengthen together) and **asymmetric** (one lengthens as the other shortens; always the higher frequency of the pair).
- **Bending** — bond angles change: in-plane **scissoring** and **rocking**; out-of-plane **wagging** and **twisting**.

Top: the three modes of water (H_2O). Bottom: bending modes of a $–CH_2–$ group (+ = out of plane, − = into plane)

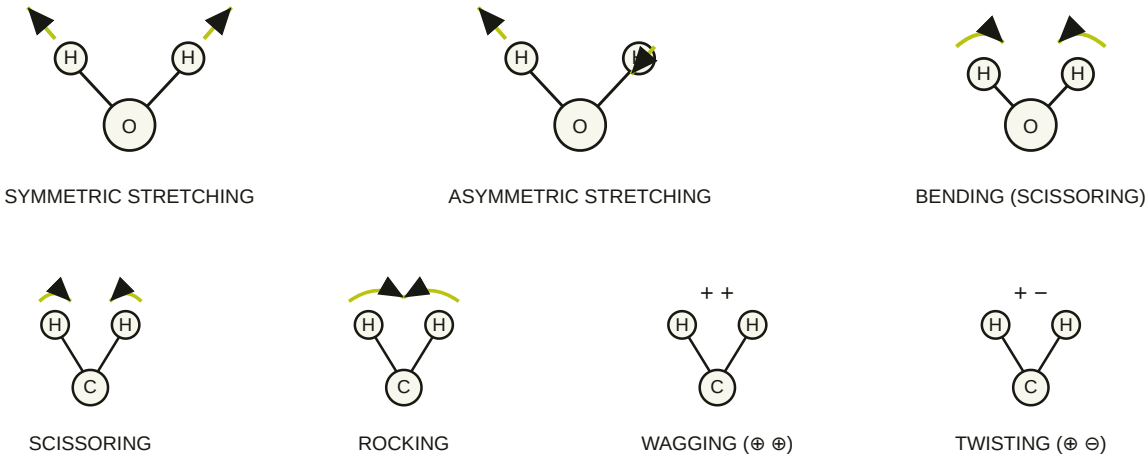


Fig. 8 — Stretching and bending vibrational modes (schematic).

WORKED EXAMPLE — COUNTING MODES

Water (non-linear, N = 3): 3N − 6 = **3 modes** — symmetric stretch 3657 cm⁻¹, bend 1595 cm⁻¹, asymmetric stretch 3756 cm⁻¹, all IR-active. Carbon dioxide (linear, N = 3): 3N − 5 = **4 modes** — symmetric stretch (Raman-active only, since the dipole moment does not change), a doubly degenerate bend (667 cm⁻¹) and the asymmetric stretch (2349 cm⁻¹).

3. Factors affecting group frequencies

- **Bond strength and atom mass** (Hooke's law): C≡C > C=C > C–C in frequency; C–H (light H) absorbs far above C–C.

- **Hydrogen bonding:** broadens and shifts X–H stretching to lower wavenumber — free O–H ≈ 3600 cm^{-1} becomes a broad band centred ≈ 3300 cm^{-1} in alcohols/water.
- **Conjugation:** delocalisation weakens a double bond slightly; a conjugated C=O absorbs ≈ 30 cm^{-1} lower than an isolated ketone (1715 → ≈ 1685 cm^{-1}).
- **Ring strain:** compressing bond angles raises stretching frequency (cyclobutanone C=O ≈ 1780 cm^{-1} vs ≈ 1715 in open-chain ketones).
- **Inductive/electronic effects:** electron-withdrawing neighbours stiffen bonds and raise frequency (acid chloride C=O ≈ 1800 cm^{-1}), donors lower it (amide C=O ≈ 1650 cm^{-1}).
- **Physical state and solvent:** hydrogen-bonding solvents and the solid state shift and broaden bands relative to dilute solution in inert solvents.

4. Sample handling and experimental techniques

- **KBr pellet:** ≈ 1 mg sample ground with dry KBr and pressed to a transparent disc — the classic solid method; moisture must be excluded (KBr is hygroscopic).
- **Nujol mull:** sample ground in liquid paraffin between salt plates — quick, but Nujol's own C–H bands mask parts of the spectrum.
- **Films and fibres:** polymer films are run directly; textile fibres are flattened, or examined by microscope-ATR.
- **Solutions:** short-path cells of NaCl/KBr windows with IR-transparent solvents (CCl_4, CS_2).
- **ATR (attenuated total reflectance):** the sample is pressed against a crystal (diamond, ZnSe); the evanescent wave samples the surface layer — the modern default for fabrics and finishes because no preparation is needed.
- **FTIR:** Fourier-transform instruments record all frequencies simultaneously through an interferometer — fast, sensitive, computer-matched against libraries, which is how fibre identification is done in practice.

5. Correlation chart (principal group frequencies)

GROUP / BOND	REGION (CM^{-1})	NOTES
O–H (H-bonded, broad)	3200–3550	Water, alcohols, cellulose — broad band
N–H	3300–3500	Amines, amides (wool/silk protein region overlaps O–H)
C–H stretch	2850–3300	Alkane ≈ 2850–2960; alkene/aromatic ≈ 3000–3100; alkyne ≈ 3300

C≡N	≈ 2240–2280	Sharp; identifies acrylic fibre
C=O	1650–1800	Position diagnoses ester (≈ 1735), ketone (≈ 1715), amide (≈ 1650), acid chloride (≈ 1800)
C=C	1620–1680	Often weak; stronger in Raman
N–H bend / amide II	≈ 1550	Protein fibres (wool, silk)
C–O	1000–1300	Strong; cellulose shows heavy absorption here
Fingerprint region	1500–400	Unique pattern per compound/fibre — library matching zone

6. Raman spectroscopy

Raman is a **scattering** technique. Intense monochromatic light (a laser) strikes the sample; most photons scatter elastically (**Rayleigh** scattering, unchanged energy), but about one in a million exchanges energy with a molecular vibration and scatters at a shifted frequency. A shift to lower frequency (molecule gained vibrational energy) gives a **Stokes** line; to higher frequency, an **anti-Stokes** line. The shift, in cm^{-1}, equals the vibration frequency directly. The selection rule differs from IR: a vibration is Raman-active if it changes the molecule's **polarisability**. Consequences:

- Vibrations silent in IR may be strong in Raman and vice versa — for centrosymmetric molecules the rule of **mutual exclusion** holds (e.g. CO_2 symmetric stretch: Raman only; asymmetric stretch: IR only). The two spectra are complementary.
- Water is a weak Raman scatterer, so aqueous samples — impractical in IR — are easy in Raman; glass containers can be used.
- Non-polar bonds (C=C, C–C, S–S) give strong Raman bands — useful for dyes, pigments and fibre backbones.

FEATURE	IR	RAMAN
Physical process	Absorption of IR radiation	Inelastic scattering of laser light
Activity requires change in	Dipole moment	Polarisability
Water as solvent	Poor (absorbs strongly)	Good (weak scatterer)
Sample cells	Salt (NaCl/KBr) windows	Glass

Strong for	Polar groups (O–H, C=O)	Non-polar bonds (C=C, S–S), symmetric modes

7. Applications

- **Fibre identification:** FTIR library matching identifies cotton, polyester (C=O ≈ 1710, aromatic bands), acrylic (C≡N ≈ 2240), nylon (amide bands), wool/silk (amide I/II) in unknown samples and blends; ATR needs no sample preparation.
- **Finish and contamination analysis:** detecting softeners, resins, oils and size residues on fabric (links directly to Topic 6 estimation).
- **Dye and pigment analysis:** Raman identifies pigments and dyes on fibres non-destructively (forensic textile work).
- **Structure studies:** crystallinity and orientation of fibres (band ratios, polarised IR/Raman), cellulose damage (carbonyl growth on ageing/oxidation), polymer degradation monitoring.
- **Quality control:** rapid verification of chemical deliveries and blend ratios in production laboratories.

8. Glossary — key terms

- **Wavenumber:** $1/\lambda$ in cm^{-1}; proportional to frequency and energy.
- **Fingerprint region:** 1500–400 cm^{-1}, uniquely characteristic of a compound.
- **Force constant k:** bond stiffness in Hooke's-law treatment of vibration.
- **Stokes / anti-Stokes lines:** Raman scattering shifted to lower / higher frequency.
- **ATR:** attenuated total reflectance — surface IR sampling against a crystal.

9. Key points

- IR absorption needs a changing dipole moment; frequency follows Hooke's law — stronger bonds and lighter atoms absorb higher.
- Modes: stretching (symmetric < asymmetric in frequency) and bending (scissoring, rocking, wagging, twisting); N atoms give $3N - 6$ (linear $3N - 5$) modes.
- Group frequencies shift predictably with H-bonding, conjugation, ring strain and inductive effects.
- Raman measures polarisability change by scattering; the two techniques are complementary (mutual exclusion in centrosymmetric molecules), and Raman tolerates water.
- In textiles the pair identify fibres, finishes, dyes and structural damage.

10. Summary

Infrared and Raman spectroscopy read the vibrational life of molecules from two directions: absorption governed by dipole change, scattering governed by polarisability change. Hooke's law explains where bands fall, vibrational modes explain their variety, and correlation charts convert them into structural evidence. With ATR-FTIR on the bench, these methods have become routine identification tools for the fibres, chemicals and finishes the rest of this course handles physically.

11. Practice

Answer in your own words; brief model answers follow.

1. Calculate the wavenumber of radiation of wavelength 6.0 μm, and state which region of the IR spectrum it falls in.
2. Using Hooke's law reasoning only, order these stretches by frequency: C–C, C=C, C≡C, and C–H vs C–C. Justify each comparison.
3. CO_2's symmetric stretch appears in Raman but not in IR, while its asymmetric stretch appears in IR but not Raman. Explain with the two selection rules.
4. An unknown fibre shows a sharp band at 2240 cm^{-1} and a strong band at ≈ 1735 cm^{-1}. Suggest the fibre and the group each band represents.
5. Why is Raman preferred for examining a dye on a wet fibre sample?

Answer key

1. $\tilde{\nu} = 1 \div (6.0 \times 10^{-4}\ cm) \approx 1667\ cm^{-1}$ — the upper fingerprint/functional-group boundary region (near C=O/C=C stretching territory).
2. C–C < C=C < C≡C (force constant rises with bond order at equal masses). C–H > C–C (much smaller reduced mass for H, outweighing similar k). Hooke: $\tilde{\nu} \propto \sqrt{(k/\mu)}$.
3. Symmetric stretching of CO_2 leaves the (zero) dipole moment unchanged → IR-inactive; but the molecule's polarisability changes as it expands/contracts → Raman-active. Asymmetric stretching creates an oscillating dipole → IR-active; the polarisability change averages out → Raman-inactive. This is mutual exclusion in a centrosymmetric molecule.
4. Acrylic fibre: 2240 cm^{-1} is the C≡N (nitrile) stretch of polyacrylonitrile; the ≈ 1735 cm^{-1} band is the C=O of the ester comonomer commonly copolymerised in acrylics.
5. Water absorbs IR so strongly that it swamps an IR spectrum, and salt cells dissolve; water is a weak Raman scatterer and the sample can sit in glass, so the dye's Raman bands are measurable directly.